



Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to guide strategic business decisions through data-driven analysis of spending patterns, customer segments, and product preferences.

Understanding Our Data Foundation

Our analysis examines 3,900 customer transactions across 18 key features, providing comprehensive insights into shopping behavior. The dataset captures customer demographics, purchase details, and behavioral patterns including subscription status, discount usage, and review ratings.

With only 37 missing values in the Review Rating column, the data quality is excellent. Key dimensions include age, gender, location, product categories, purchase amounts, seasonal trends, and shipping preferences.

Dataset Snapshot

- 3,900 transactions analyzed
- 18 feature dimensions
- 4 product categories
- 50 unique locations
- 99% data completeness

Data Preparation & Engineering

We implemented a comprehensive data preparation pipeline in Python to ensure analysis-ready data quality and consistency.

01

Data Loading & Exploration

Imported dataset using pandas and performed initial exploration with `info()` and `describe()` functions to understand structure and statistics.

02

Missing Data Handling

Imputed 37 missing Review Rating values using median ratings by product category to maintain data integrity.

03

Feature Engineering

Created `age_group` bins and `purchase_frequency_days` columns. Standardized column names to `snake_case` format.

04

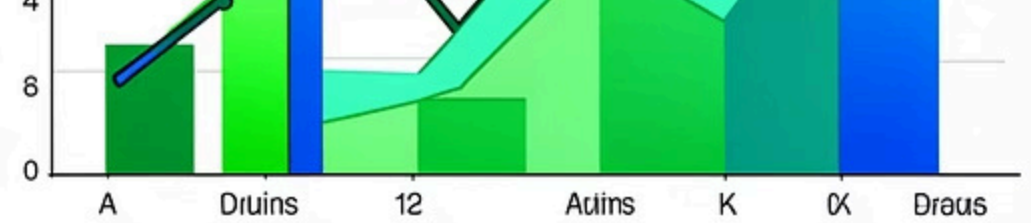
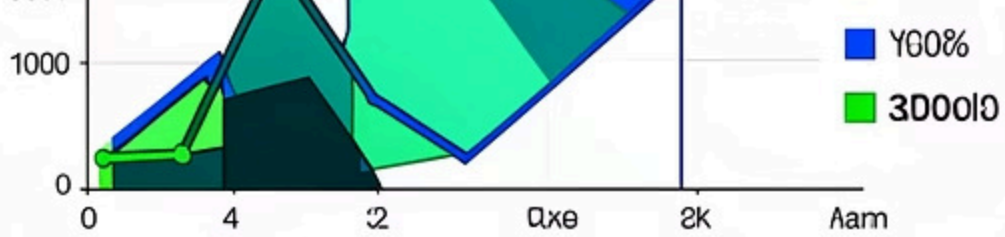
Data Consistency

Verified redundancy between `discount_applied` and `promo_code_used` fields, dropping the latter for cleaner analysis.

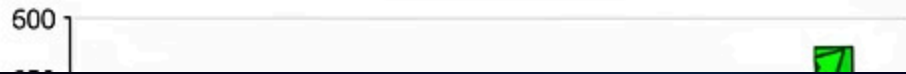
05

Database Integration

Connected to PostgreSQL and loaded cleaned DataFrame for structured SQL analysis and business intelligence.



Customer Purchase Pathens



Key Statistical Insights

\$60

Average Purchase

Mean transaction value across
all customers

3.75

Review Rating

Average customer satisfaction
score

44

Median Age

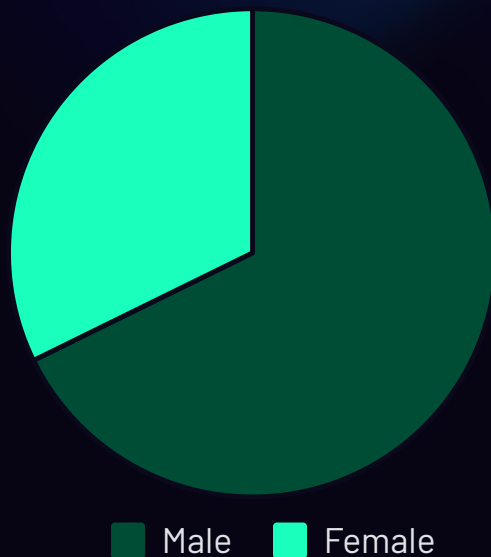
Typical customer age profile

25

Previous Purchases

Average customer purchase
history

Revenue Patterns & Customer Segments



Gender Revenue Distribution

Male customers generate 68% of total revenue at \$157,890, significantly outpacing female customers at \$75,191. This 2:1 ratio suggests opportunities for targeted marketing campaigns to balance customer acquisition and engagement across gender segments.

Understanding these patterns enables more effective resource allocation and personalized shopping experiences.

Product Performance & Customer Preferences



Gloves

Top-rated product with 3.86 average rating, leading customer satisfaction scores.



Sandals

Second highest rating at 3.84, popular in footwear category with strong reviews.



Boots

Rated 3.82, consistently high performer in customer satisfaction metrics.



Hat

50% discount rate leader with 3.80 rating, highest promotion dependency.

Shipping & Discount Behavior Analysis

Shipping Type Impact

Express shipping customers spend \$60.48 on average, slightly higher than Standard shipping at \$58.46. This \$2 difference suggests premium shipping attracts marginally higher-value transactions.

Express

\$60.48 avg

Standard

\$58.46 avg

Smart Discount Users

839 customers used discounts while spending above the \$60 average, demonstrating that promotions don't always reduce transaction values. These savvy shoppers represent high-value customers who respond to incentives.



Customer Segmentation & Loyalty Patterns



Loyal Customers

3,116 customers (80%) with 5+ purchases form our core base, driving consistent revenue.



Returning Customers

701 customers (18%) with 2-4 purchases show growth potential for loyalty programs.

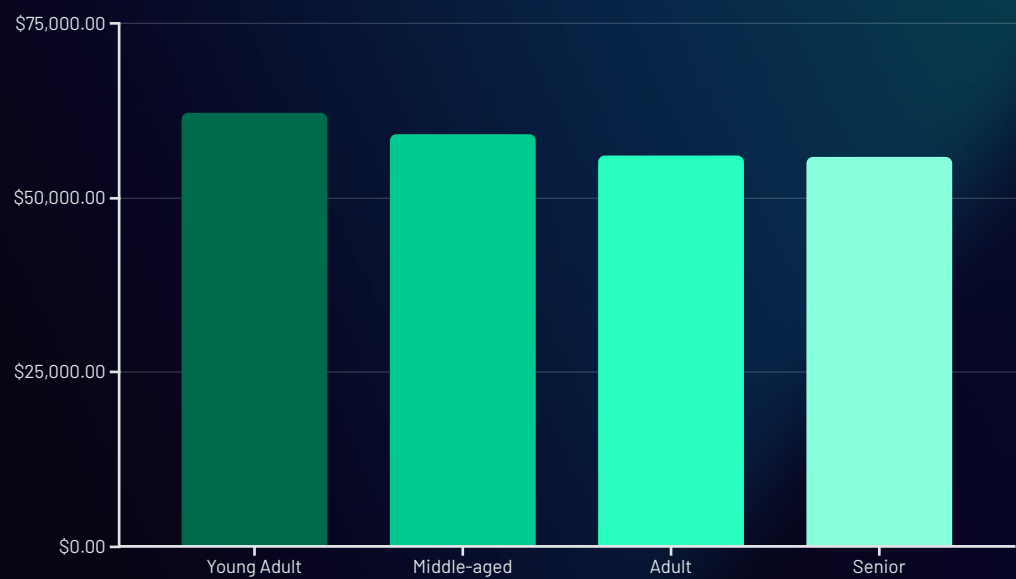


New Customers

83 customers (2%) with single purchases represent acquisition opportunities.

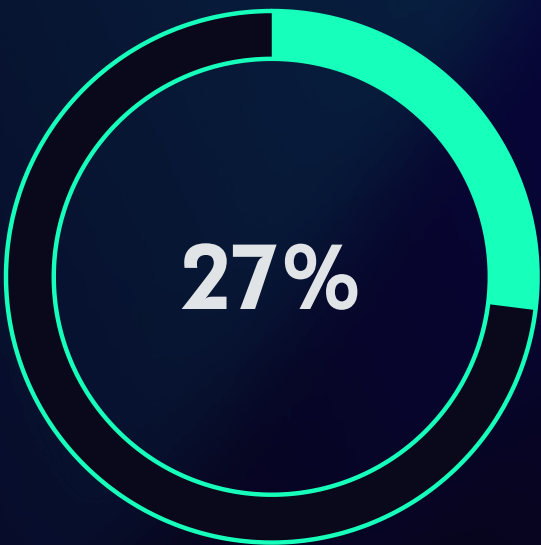
Among repeat buyers with 5+ purchases, 958 are subscribers while 2,518 are not—indicating significant subscription conversion potential within our most valuable customer segment.

Subscription & Age Group Revenue Analysis

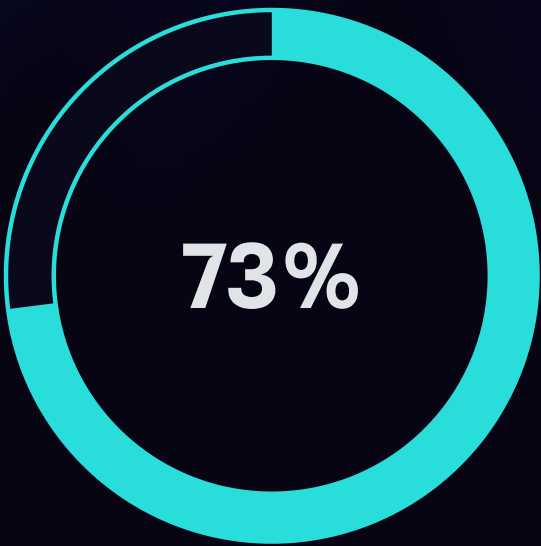


Revenue Distribution Insights

Young Adults lead revenue generation at \$62,143, followed closely by Middle-aged customers at \$59,197. The relatively even distribution across age groups suggests broad market appeal.



Subscription rate across all customers



Non-subscribers with conversion potential

Subscribers and non-subscribers show similar average spend (\$59.49 vs \$59.87), but non-subscribers generate 2.7x more total revenue due to volume.

Actionable Business Recommendations

Boost Subscription Conversion

With 73% non-subscribers and 2,518 repeat buyers not subscribed, promote exclusive benefits and incentives to convert high-value customers into subscribers.

Implement Loyalty Programs

Reward the 701 returning customers to accelerate their transition into the loyal segment, increasing lifetime value and purchase frequency.

Optimize Discount Strategy

Balance promotional activities carefully—839 customers spend above average despite discounts, but products like Hats show 50% discount dependency requiring margin review.

Highlight Top Products

Feature high-rated products (Gloves 3.86, Sandals 3.84, Boots 3.82) prominently in marketing campaigns to leverage proven customer satisfaction.

Target High-Value Segments

Focus marketing efforts on Young Adults (\$62K revenue), male customers (68% of revenue), and express shipping users who demonstrate higher transaction values.